INFOTACT SOLUTIONS

EchoChain

Circular Economy & Secondary Market Lifecycle Analytics

From Marketplace Data to Lifecycle Insights

Organization	Infotact Solutions
Project Duration	10 August – 7 September
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Technology Stack	Python • Scrapy • Databricks • Delta Lake • PySpark • Power BI • SQL • Git & GitHub

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1. Introduction

Project Overview

EchoChain is a lifecycle analytics project that connects secondary-market electronics listings with product reference, Bill of Materials (BOM), and warranty information. The goal is to bring these data sources together so marketplace value, component health, warranty failures, and lifecycle indicators can be explored in one analytical view.

Problem Statement & Use Case

Manufacturers track products closely until sale; after sale, lifecycle data becomes a blind spot, limiting measurement of environmental impact, landfill diversion, and refurbishment opportunities. EchoChain joins scraped secondary-market listings with BOM and warranty data. A product with a high Circularity Score, frequent warranty failures, and strong resale value can be investigated as a potential buy-back or refurbishment opportunity.

Objectives

• Collect and prepare marketplace listing data and supporting SKU, BOM, and warranty datasets.

• Organize raw, cleaned, and analytics-ready data through Bronze, Silver, and Gold layers.

• Use SKU extraction and fuzzy matching to link marketplace titles with official product references.

• Create lifecycle metrics and present them through interactive Power BI pages.

Technology Stack

LAYER	TECHNOLOGY	ROLE
Collection	Python, Scrapy	Marketplace listing acquisition and extraction
Storage	Databricks, Delta Lake	Layered data storage and table management
Processing	PySpark	Cleaning, SKU extraction, matching, transformations
Visualization	Power BI	Lifecycle KPIs, trends, drill-downs, anomaly views
Collaboration	Git, GitHub	Source control and project organization

2. System Architecture

The architecture moves marketplace and supporting product data through Databricks. Bronze retains ingested raw data, Silver provides cleaned and standardized data, SKU matching links listings to official products, and Gold contains integrated lifecycle information used by Power BI.

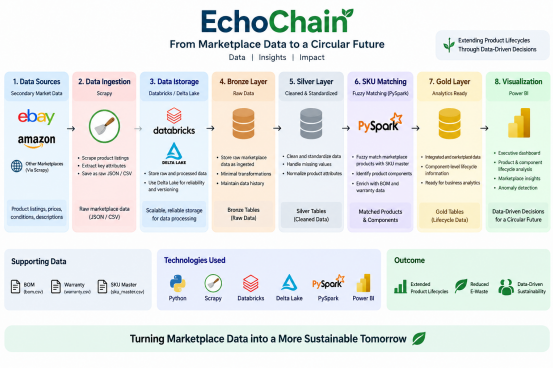


Figure 2. Marketplace-to-Power BI workflow

3. Project Implementation

Step 1 — Marketplace and Supporting Data

The project uses marketplace listing records together with three supporting datasets: SKU Master, BOM, and warranty data. These datasets provide product reference and component-level context. BOM and warranty records are project datasets for this analysis and should not be interpreted as verified records from a real manufacturer.

Step 2 — Data Ingestion and Bronze Layer

Marketplace records are ingested into the Databricks workflow and stored in Bronze. This layer retains the raw ingested representation and supports traceability before further transformations.

Step 3 — Silver Transformation

The Bronze-to-Silver step cleans and standardizes fields, handles missing values where defined, and normalizes product attributes so records are more consistent for downstream processing.

Step 4 — SKU Extraction and Fuzzy Matching

The pipeline extracts product/model information from marketplace titles and uses fuzzy matching in PySpark to connect listings to the official SKU master. This helps address differences between marketplace title text and standardized product identifiers. Matched records are enriched with BOM and warranty information.

Step 5 — Gold Lifecycle Dataset

The final transformation creates the Gold lifecycle dataset used by Power BI. It combines marketplace listing information with official SKU, component, manufacturing-cost, and warranty attributes, allowing market data to be analyzed alongside component-level lifecycle indicators.

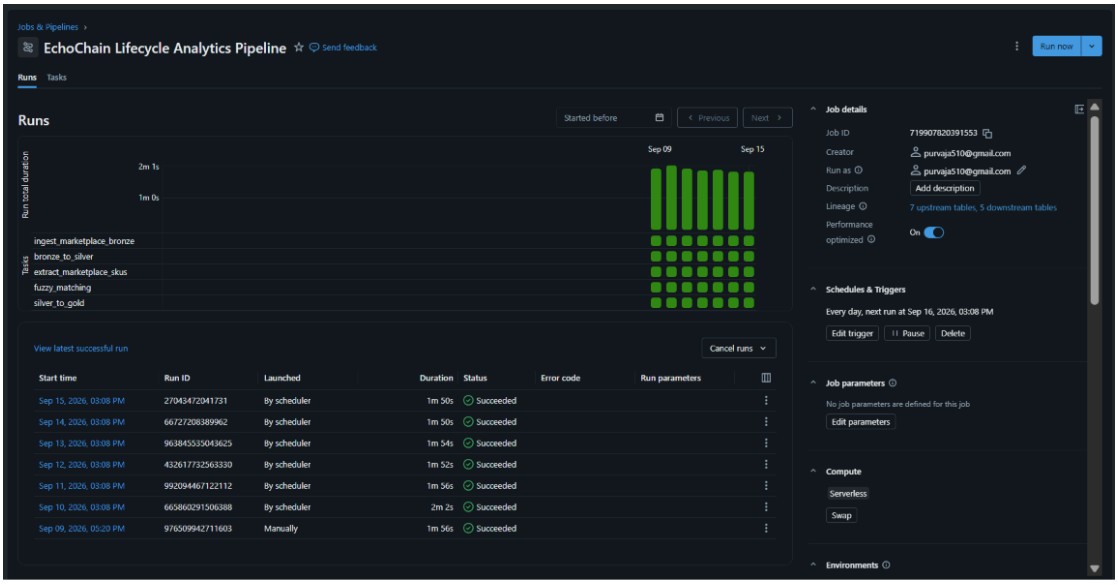


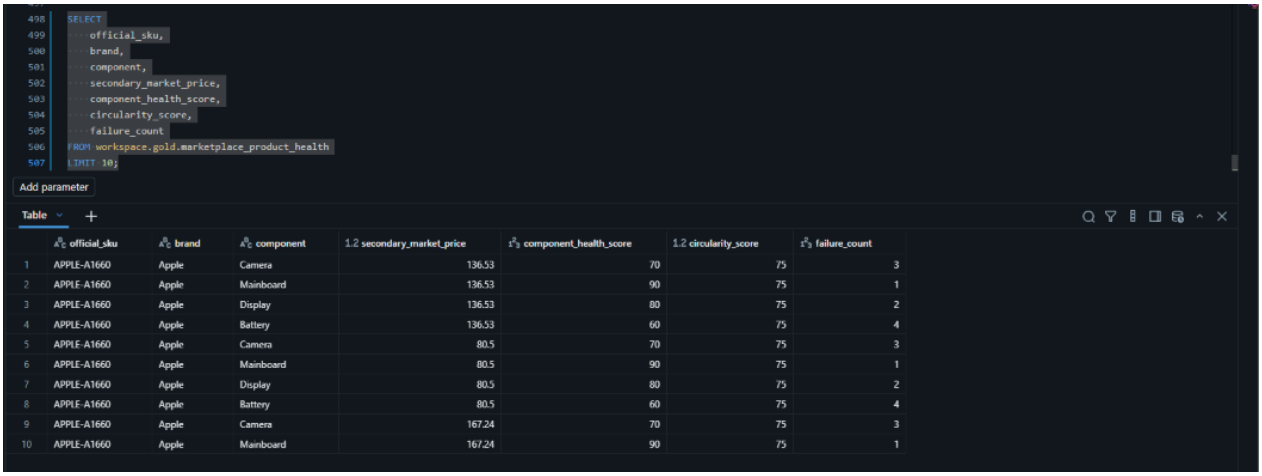
Figure 3. Databricks Lifecycle Analytics Pipeline: five dependent tasks and successful scheduled runs

Task sequence: ingest_marketplace_bronze → bronze_to_silver → extract_marketplace_skus → fuzzy_matching → silver_to_gold.

4. Gold Dataset & Databricks Automation

Gold Lifecycle Dataset

The table **workspace.gold.marketplace_product_health** is the analytics-ready output of the pipeline. The sample below shows official SKU, brand, component, secondary-market price, component health score, circularity score, and warranty failure count. Gold data is used as the main source for the Power BI lifecycle views.



```
SELECT
    official_sku,
    brand,
    component,
    secondary_market_price,
    component_health_score,
    circularity_score,
    failure_count
FROM workspace.gold.marketplace_product_health
LIMIT 10;
```

Add parameter

Table +

	official_sku	brand	component	secondary_market_price	component_health_score	circularity_score	failure_count
1	APPLE-A1660	Apple	Camera	136.53	70	75	3
2	APPLE-A1660	Apple	Mainboard	136.53	90	75	1
3	APPLE-A1660	Apple	Display	136.53	80	75	2
4	APPLE-A1660	Apple	Battery	136.53	60	75	4
5	APPLE-A1660	Apple	Camera	80.5	70	75	3
6	APPLE-A1660	Apple	Mainboard	80.5	90	75	1
7	APPLE-A1660	Apple	Display	80.5	80	75	2
8	APPLE-A1660	Apple	Battery	80.5	60	75	4
9	APPLE-A1660	Apple	Camera	167.24	70	75	3
10	APPLE-A1660	Apple	Mainboard	167.24	90	75	1

Figure 4. Gold lifecycle dataset — SQL query and sample records

Pipeline Orchestration

The Databricks Job contains five dependent Python script tasks covering Bronze ingestion, Silver transformation, SKU extraction, fuzzy matching, and Gold generation. The supplied job screenshot shows successful runs and a daily schedule. Performance optimization is enabled in the job settings.

Data Validation

Validation checks support confidence in the data used for analysis. The project's reported Gold checks include row and SKU counts, matched listing counts, null/range checks, and lifecycle metric summaries. The table is at listing-component grain, so one marketplace listing can appear in multiple rows—one per associated component.

Interpreting Depreciation

The depreciation measure compares secondary-market price with the manufacturing-cost baseline available from BOM data. A negative value means the marketplace price is above that baseline; it is a market premium relative to manufacturing cost, not proof that a product appreciated against its original retail price.

5. Dashboard & Business Insights

Power BI provides three views: Executive Lifecycle Overview, Product & Component Lifecycle Analysis, and Marketplace & Anomaly Analysis. Filters for brand, condition, component, and official SKU help users narrow the analysis.

Executive Lifecycle Overview

The overview presents average circularity score (78.04), average secondary-market price (420.49), total warranty failures (5.05K), and average depreciation versus manufacturing cost (-2.58%). The brand comparison combines average marketplace price and circularity score. Other visuals compare warranty failures with market value and provide a brand-to-SKU-to-component drill-down.

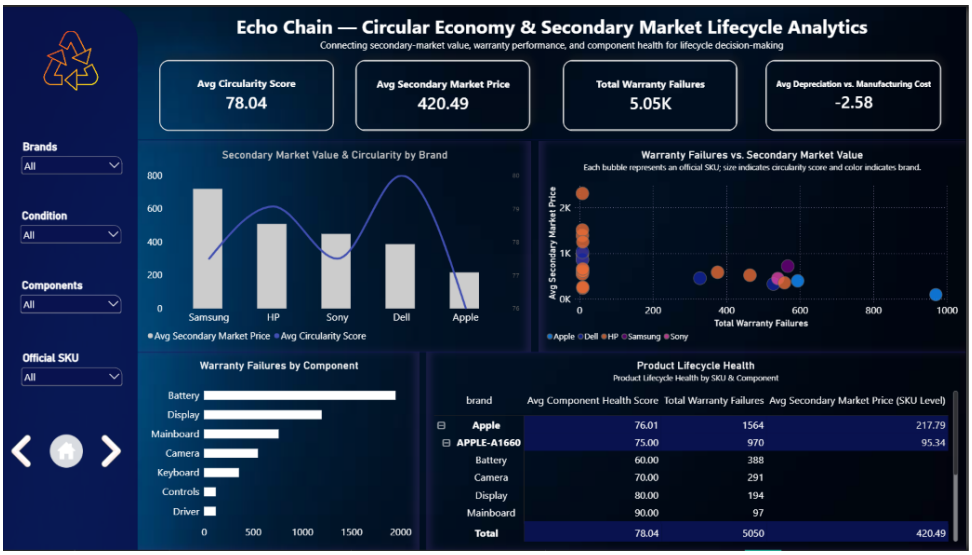


Figure 5. Executive Lifecycle Overview

Product & Component Lifecycle Analysis

This page focuses on component health and warranty performance. It highlights average component health, total warranty failures, circularity score, components requiring attention, and a component-level breakdown. These visuals help identify where further investigation may be useful; they do not by themselves establish the cause of a failure.

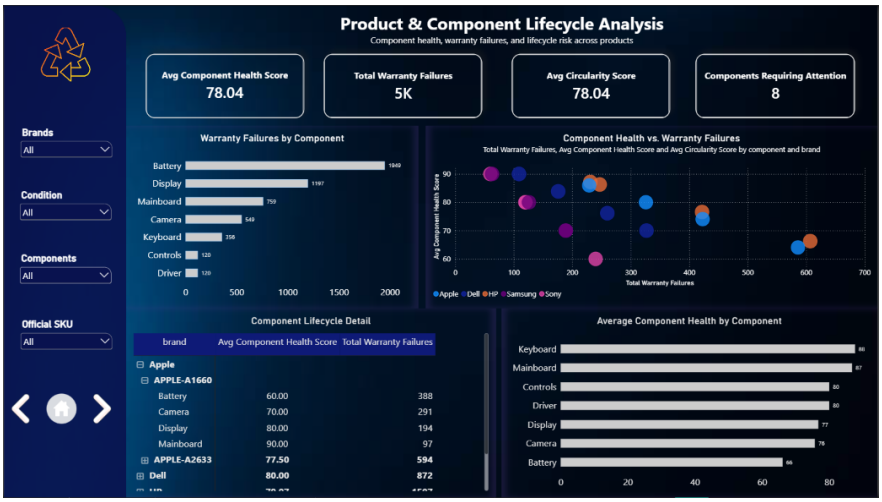


Figure 6. Product & Component Lifecycle Analysis

5. Dashboard & Business Insights (continued)

Marketplace & Anomaly Analysis

This page examines secondary-market price patterns, depreciation relative to manufacturing cost, and unusual listings. The dashboard includes an anomaly detail table for investigation. IQR-based screening flags unusual prices for review; flagged records are not automatically treated as invalid or removed.

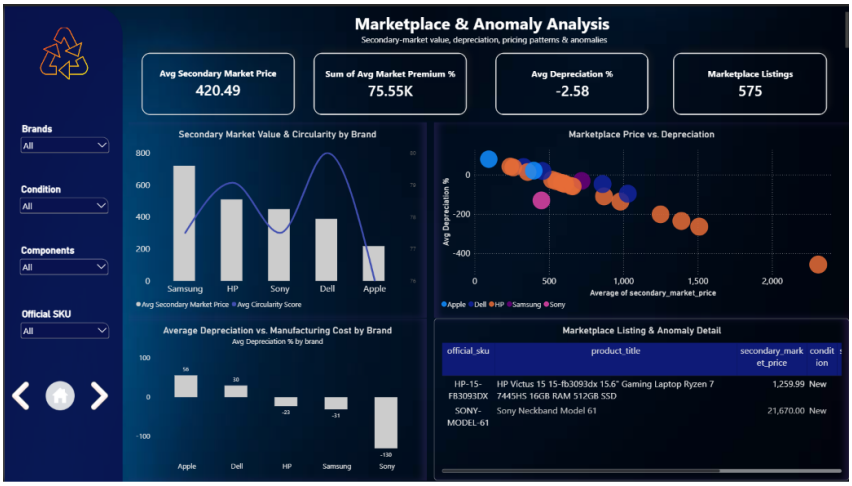


Figure 7. Marketplace & Anomaly Analysis

6. Version Control & Project Structure

GitHub organizes the project into folders for data, Databricks assets, documentation, images, Power BI, PySpark, Scrapy, and validation scripts. This keeps data engineering code, dashboard materials, and supporting documentation together for review and collaboration.

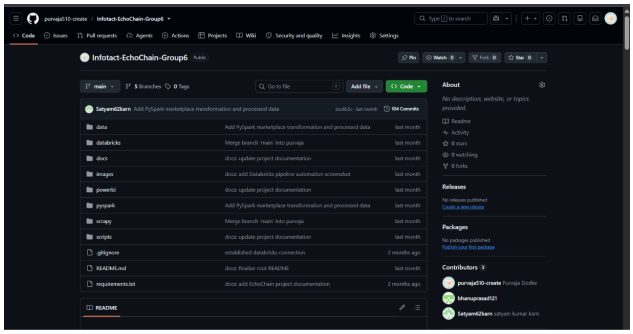


Figure 8. EchoChain GitHub repository and project folders

7. Project Highlights

• Layered workflow from raw marketplace records to a Gold lifecycle dataset.

• Fuzzy SKU matching to connect marketplace listings with official product references.

• Integration of marketplace, BOM, SKU master, and warranty information.

• Scheduled Databricks orchestration with successful runs shown in the supplied screenshot.

• Power BI views for executive indicators, component analysis, and marketplace anomalies.

Challenges & Considerations

Marketplace titles can differ from standardized SKU descriptions, so matching requires careful review. Unusual marketplace prices should be investigated rather than automatically deleted. Depreciation is based on manufacturing cost in the current data, not a dedicated original retail-price field.

8. Conclusion & Future Scope

EchoChain demonstrates an end-to-end analytics workflow that transforms marketplace and supporting product data into lifecycle indicators. The Gold dataset supports analysis of market value, component health, and warranty failures through Power BI. Future improvements could include verified product and retail-price data, expanded validation, match-confidence review, and further automation of data-quality checks.